Identifying Undiagnosed Type 2 Diabetics Using CatBoost Modeling

Tyson Johnson, Alizeh Murtaza, Nithila Neethi Devan, Tariq Abdulhak, Dr. Mohamed Adel Slamani

Dept. of Computational & Data Sciences — CDS-492 Capstone in Data Science

COLLEGE OF SCIENCE
George Mason University®

Introduction

Type 2 diabetes (T2D) is a chronic condition in which the body loses the ability to regulate blood sugar effectively. Over time, elevated blood sugar damages blood vessels and organs, leading to heart disease, kidney failure, nerve damage, and vision loss. Early detection is critical because lifestyle changes and medication can slow or prevent these complications, but only if the condition is caught in time.

The International Diabetes Federation estimates that 44.7% of diabetes cases worldwide remain undiagnosed [1]. This is a staggering gap: nearly half of all people living with diabetes do not know they have it. Routine health checkups already measure the key indicators, including fasting blood glucose, body mass index (BMI), blood pressure, and cholesterol, but a doctor reviewing one test at a time may miss subtle combinations of warning signs.

Machine learning offers a different approach: instead of looking at one measurement at a time, a model can analyze dozens of variables simultaneously and learn which patterns of results tend to appear together in patients with diabetes. The model then assigns each patient a risk score. This project asks: can we build an accurate, automated screening tool from standard checkup data that flags likely undiagnosed T2D patients for follow-up testing? We used publicly available data from South Korea's National Health Insurance Service (NHIS) [2], containing one million anonymized patient records, to train and compare eight different model types.

We selected the South Korean NHIS dataset for several reasons. South Korean citizens have access to free health examinations every two years through their national insurance system, and these exams include comprehensive blood work (fasting glucose, cholesterol panels, liver enzymes, and more). Because participation is widespread and standardized, the NHIS dataset is one of the largest publicly available collections of real-world health screening data, containing approximately one million anonymized records from 2024. Its structure closely mirrors what clinics collect worldwide, making findings potentially transferable to other healthcare systems. We trained and compared eight different model types on this data to determine which approach best balances accuracy with practical usability.

Methodology

The NHIS dataset we selected includes demographics (age, sex, region), body measurements (height, weight, waist circumference), vital signs (systolic and diastolic blood pressure), and laboratory results (fasting blood glucose, total cholesterol, HDL and LDL cholesterol, triglycerides, liver enzymes AST/ALT/GGT, hemoglobin, serum creatinine, and urinalysis).

Since the dataset does not include a formal diabetes diagnosis, we defined a binary diabetes label using fasting plasma glucose (FPG): individuals with FPG ≥ 126 mg/dL were labeled positive, following the American Diabetes Association diagnostic threshold. From the raw measurements, we engineered 40 input features for the models, including derived variables like BMI, waist-to-height ratio, pulse pressure, lipid ratios (e.g., triglycerides/HDL), log-transformed lab values to normalize skewed distributions, and indicators for whether optional tests like the lipid panel were completed or missing.

Figure 1 shows the distribution of the FPG ranges within the dataset as diabetic category labels. To provide background on how many patients were close to diabetic, patients with 100 ≤ FPG < 126 were additionally labeled pre-diabetic.

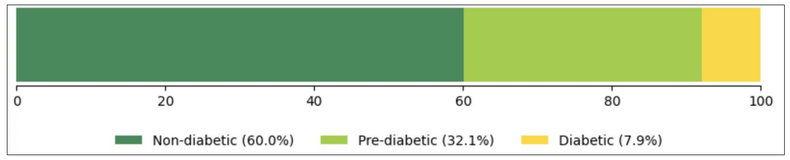

Fig. 1. Distribution of prevalence of diabetics in dataset.

The data were split into three sets: training (60%), validation (20%), and test (20%). The training set teaches the model to recognize patterns; the validation set is used during training to prevent overfitting (where a model memorizes the training data instead of learning general patterns); and the test set is held out entirely to give an unbiased measure of final performance. All eight models were trained and evaluated on the exact same splits to ensure a fair comparison. We applied class weighting so the models do not simply learn to predict "no diabetes" for everyone.

We tested eight different models to determine which approach works best for this type of data. These ranged from simple, interpretable methods (logistic regression and decision trees) to more complex approaches (neural networks and gradient-boosted ensembles). Our primary model, CatBoost, is a gradient-boosted decision tree ensemble, meaning it builds many small decision trees sequentially, with each new tree improving on earlier errors.

We trained three CatBoost variants using progressively larger sets of input variables (15, 26, and 40 features) to measure how much each category of health data contributes to prediction accuracy.

This three-tier design reflects a practical reality of the dataset: lipid panel tests (total cholesterol, HDL, LDL, triglycerides) are optional in the NHIS health checkup, so many patients do not have those results. If a model depends heavily on cholesterol values, it will not work for patients who skipped those tests or in clinics that do not routinely order them. By training separate models with and without lipid features, we can measure exactly how much accuracy is lost when those variables are unavailable — and whether a model without them is still useful.

We also trained a logistic regression, a support vector machine (SVM), a single decision tree, and two types of artificial neural networks (an sklearn MLPClassifier and a PyTorch tabular network). Finally, rather than using a default 50/50 classification cutoff, we selected a threshold targeting ≥85% recall. This means the model is tuned to catch at least 85% of true diabetes cases, accepting more false alarms in exchange for fewer missed diagnoses, which is the appropriate trade-off for a screening tool.

To understand our results, it helps to know the four possible outcomes when a model classifies a patient. A true positive correctly identifies someone with diabetes for follow-up. A false positive incorrectly flags a healthy person, resulting in an unnecessary but harmless follow-up blood test. A false negative is the most dangerous outcome: the model misses someone who actually has diabetes, and they go untreated. A true negative correctly clears a healthy person. Sensitivity (also called recall) is the percentage of actual diabetes cases the model catches — at ≥85%, out of every 100 people who truly have diabetes, the model correctly flags at least 85.

ROC-AUC measures how well the model ranks risk. If given one patient with diabetes and one without, how often does the model correctly rank the diabetic patient as higher risk? A perfect model scores 1.0, while random guessing scores 0.5. We calibrated the model's predicted risk scores so that a reported 10% risk corresponds to about 10 out of 100 similar patients actually having diabetes.

Results

Figure 2 compares all eight models by ROC-AUC. The three best-performing models were CatBoost Full (0.817), the PyTorch neural network (0.816), and the sklearn ANN (0.812), all achieving nearly identical scores despite being fundamentally different model architectures. Simpler baselines like logistic regression and the linear SVM still performed well at 0.792, outperforming the 15-feature CatBoost model (0.775), which shows that having the right input data matters more than having a complex model.

The feature set comparison revealed a key practical finding: adding standard laboratory blood tests (*core + labs*) to the basic screening (*core only*) measurements produced the largest accuracy gain (AUC improved from 0.775 to 0.808), while adding the optional lipid panel (*full model*) on top of that contributed a much smaller improvement (+0.009). This means a strong screening model can be deployed even in settings where cholesterol tests are not available.

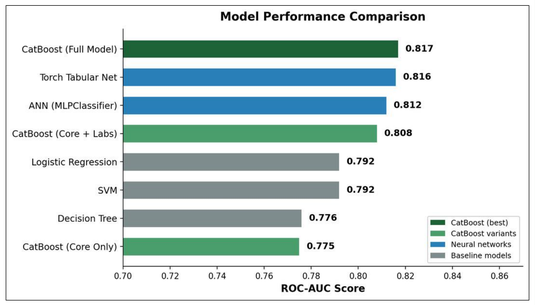

Fig. 2. ROC-AUC values for all tested models.

Figure 3 shows the full CatBoost model achieves high sensitivity of 0.852 with a specificity of 0.625, though 34.6% of total patients in the dataset were recommended to receive screening tests despite having FPG < 126. **Figure 4** uses SHAP analysis to reveal which variables drive the CatBoost model's predictions. Age was the dominant predictor, followed by the AST/ALT liver enzyme ratio, GGT, and LDL cholesterol. Notably, although the top three are all measurements collected during a routine health checkup, LDL cholesterol within the top five serves as evidence that a lipid panel test contributes meaningfully in improving the quality of predictions the CatBoost model is able to make.

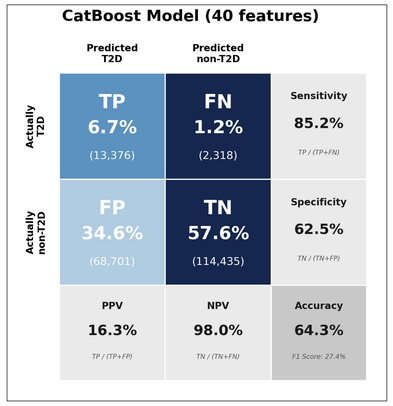

Fig. 3. CatBoost (full model) confusion matrix evaluated on the held-out test set.

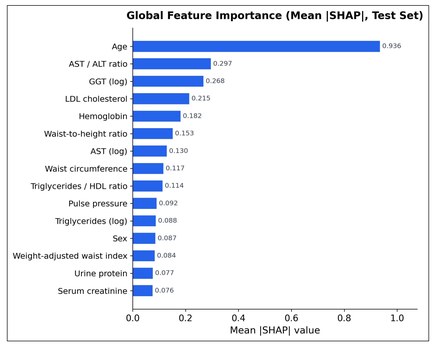

Fig. 4. Global feature importance by mean |SHAP| on the test set. Top features include age, AST/ALT ratio, GGT, LDL cholesterol, hemoglobin, waist-to-height ratio, AST, waist circumference, triglycerides/HDL ratio, and sex.

Discussion

Our best model, CatBoost, performed just as well as more complex neural network approaches (all within 0.005 AUC of each other) while being faster to train and easier to interpret. This is a meaningful finding: for structured health data like checkup records, a simpler model can match the accuracy of deep learning. This makes CatBoost more practical for real-world deployment, where clinics need tools that are transparent and maintainable. CatBoost also handles missing data natively, which is a practical advantage since real-world health records frequently have gaps when patients skip optional tests. Rather than simply ignoring or averaging over missing values, CatBoost learns from the pattern of missingness itself, and patients that skipped the lipid panel may correlate with factors such as age. This means the absence of a test result can carry predictive signal, not just noise.

The SHAP analysis (**Figure 4**) reveals which health measurements drive the model's predictions. Age, liver enzyme ratios, and cholesterol-related markers dominate, while BMI-related effects are likely captured through correlated measures such as waist circumference. Importantly, these are all collected in routine checkups, meaning no additional testing is required for deployment.

We deliberately tuned our model to catch at least 85% of true diabetes cases, even though this means some healthy patients will be incorrectly flagged. In a screening context, this trade-off makes sense: a false alarm simply leads to a follow-up blood test, while a missed diagnosis means a patient continues living with untreated diabetes and accumulating organ damage.

This study has several limitations worth noting. First, the dataset comes from a single country (South Korea), and differences in diet, genetics, and healthcare practices mean the model may not perform identically in other populations. Second, our diabetes label is based on a single fasting glucose reading rather than a clinical diagnosis, so some patients may be mislabeled. Third, the cross-sectional design means the model identifies patients who likely already have diabetes based on current markers — it does not predict who will develop diabetes in the future. Finally, the high false positive rate (34.6%) means roughly one in three flagged patients would not actually have diabetes, which has implications for follow-up testing costs and patient anxiety. Future work could validate the model on multi-national datasets, incorporate longitudinal data to enable predictive screening, and explore whether combining the model's risk score with existing clinical guidelines improves diagnostic workflows.

References

[1] International Diabetes Federation. (2021). *Diabetes now affects one in 10 adults worldwide.* https://idf.org/news/diabetes-now-affects-one-in-10-adults-worldwide/

[2] National Health Insurance Service. (2025). *National Health Insurance Service — Health checkup information* [Data set]. Public Data Portal. https://www.data.go.kr/en/data/15007122/fileData.do

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